Example 7.19

Evaluate the following limits:

(i) $\lim_{x \to 5} \frac{x^3 - 25}{x - 3}$ (ii) $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$

Solution:

(i) $\lim_{x \to 5} \frac{x^3 - 25}{x - 3} = \frac{5^3 - 25}{5 - 3} = \frac{100}{2} = 50$

(ii) $\lim_{x \to 2} \frac{x^2 - 4}{x - 2} = \frac{0}{0}$, an indeterminate form. Therefore first factorise and simplify and then apply the same limit to get the limiting value.

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2} = \lim_{x \to 2} \frac{(x + 2)(x - 2)}{x - 2}$$

$$= \lim_{x \to 2} (x + 2)$$

$$= 2+2 = 4$$

$$\therefore \quad \lim_{x \to 2} \frac{x^2 - 4}{x - 2} = 4$$

Example 7.20

Find the derivative of the following with respect to x:

(i) x^{15}-10 (ii) x^3+3x^2-6 (iii) x^4e^x (iv) $\frac{x^2 - 1}{x + 3}$

Solution:

(i) Let $y = x^{15}$-10

$$\frac{dy}{dx} = 15x^{15-1} - 0$$

$$= 15x^{14}$$

(ii) Let $y = x^3+3x^2$-6

$$\frac{dy}{dx} = 3x^2+3(2x)-0$$

$$= 3x^2+6x$$

(iii) Let $y = x^4e^x$

This is of the type

$$[uv]' = u'v + uv'$$